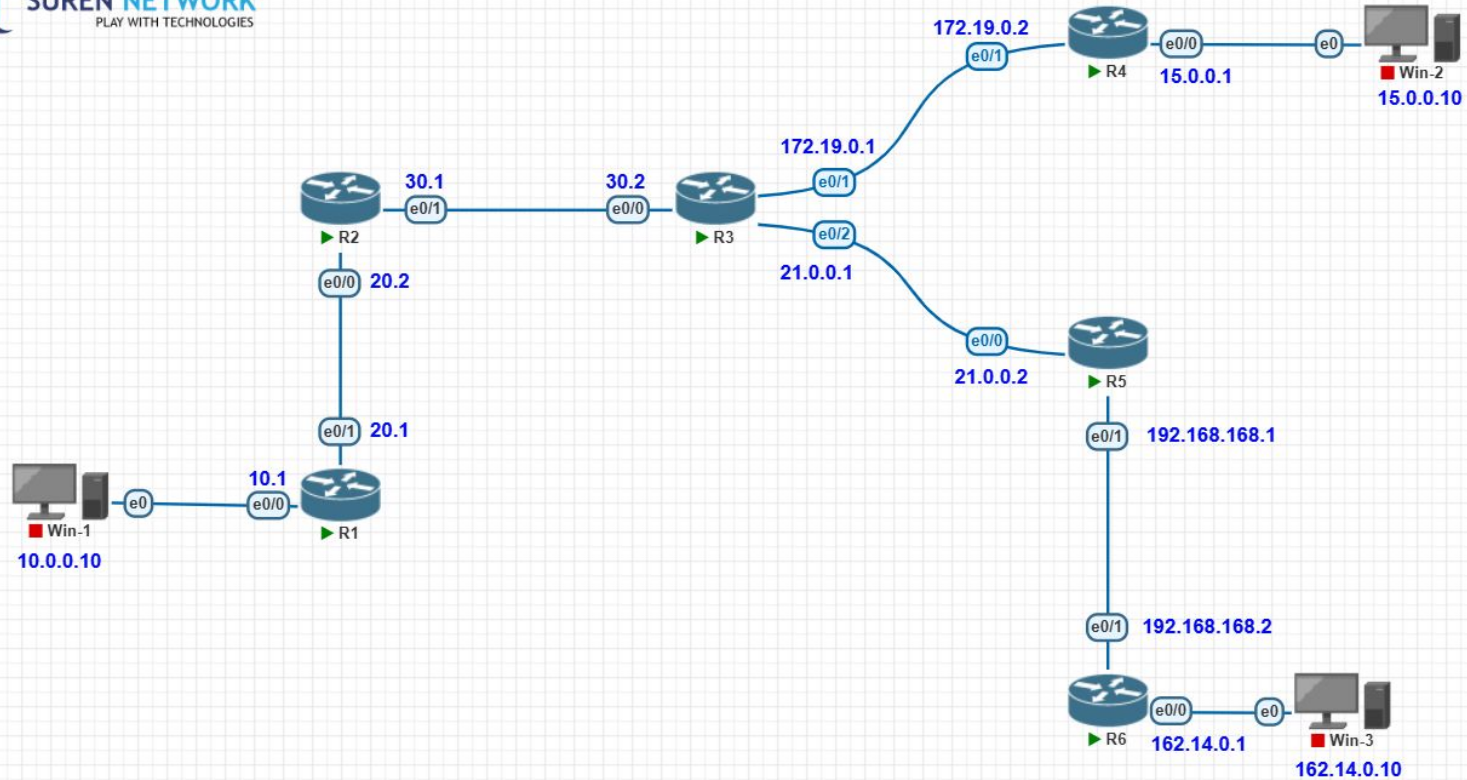




i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 10.0.0.0/8 is directly connected, Ethernet0/0
L 10.0.0.1/32 is directly connected, Ethernet0/0
S 15.0.0.0/8 [1/0] via 20.0.0.2
20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 20.0.0.0/8 is directly connected, Ethernet0/1
L 20.0.0.1/32 is directly connected, Ethernet0/1
S 21.0.0.0/8 [1/0] via 20.0.0.2
S 30.0.0.0/8 [1/0] via 20.0.0.2
S 162.14.0.0/16 [1/0] via 20.0.0.2
S 172.19.0.0/16 [1/0] via 20.0.0.2
S 192.168.168.0/24 [1/0] via 20.0.0.2

R1#

R1#

R1#ping 192.168.168.0

Type escape sequence to abort.



R2

```
R2#sh ip route
```

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP  
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area  
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2  
E1 - OSPF external type 1, E2 - OSPF external type 2  
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2  
ia - IS-IS inter area, * - candidate default, U - per-user static route  
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP  
a - application route  
+ - replicated route, % - next hop override, p - overrides from PfR
```

```
Gateway of last resort is not set
```

```
S    10.0.0.0/8 [1/0] via 20.0.0.1  
S    15.0.0.0/8 [1/0] via 30.0.0.2  
     20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks  
C     20.0.0.0/8 is directly connected, Ethernet0/0  
L     20.0.0.2/32 is directly connected, Ethernet0/0  
S    21.0.0.0/8 [1/0] via 30.0.0.2  
     30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks  
C     30.0.0.0/8 is directly connected, Ethernet0/1  
L     30.0.0.1/32 is directly connected, Ethernet0/1  
S    172.19.0.0/16 [1/0] via 30.0.0.2
```

```
R2#
```

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

```
S    10.0.0.0/8 [1/0] via 30.0.0.1
S    15.0.0.0/8 [1/0] via 172.19.0.2
S    20.0.0.0/8 [1/0] via 30.0.0.1
    21.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C      21.0.0.0/8 is directly connected, Ethernet0/2
L      21.0.0.1/32 is directly connected, Ethernet0/2
    30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C      30.0.0.0/8 is directly connected, Ethernet0/0
L      30.0.0.2/32 is directly connected, Ethernet0/0
    172.19.0.0/16 is variably subnetted, 2 subnets, 2 masks
C      172.19.0.0/16 is directly connected, Ethernet0/1
L      172.19.0.1/32 is directly connected, Ethernet0/1
```

R3#


```
R4#sh ip route
```

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP  
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area  
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2  
E1 - OSPF external type 1, E2 - OSPF external type 2  
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2  
ia - IS-IS inter area, * - candidate default, U - per-user static route  
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP  
a - application route  
+ - replicated route, % - next hop override, p - overrides from PfR
```

```
Gateway of last resort is not set
```

```
S    10.0.0.0/8 [1/0] via 172.19.0.1  
    15.0.0.0/8 is variably subnetted, 2 subnets, 2 masks  
C      15.0.0.0/8 is directly connected, Ethernet0/1  
L      15.0.0.1/32 is directly connected, Ethernet0/1  
S    20.0.0.0/8 [1/0] via 172.19.0.1  
S    30.0.0.0/8 [1/0] via 172.19.0.1  
    172.19.0.0/16 is variably subnetted, 2 subnets, 2 masks  
C      172.19.0.0/16 is directly connected, Ethernet0/0  
L      172.19.0.2/32 is directly connected, Ethernet0/0
```

```
R4#
```



R5

```
R5#sh ip route
```

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP  
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area  
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2  
E1 - OSPF external type 1, E2 - OSPF external type 2  
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2  
ia - IS-IS inter area, * - candidate default, U - per-user static route  
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP  
a - application route  
+ - replicated route, % - next hop override, p - overrides from PfR
```

```
Gateway of last resort is not set
```

```
S      10.0.0.0/8 [1/0] via 21.0.0.1  
S      20.0.0.0/8 [1/0] via 21.0.0.1  
      21.0.0.0/8 is variably subnetted, 2 subnets, 2 masks  
C      21.0.0.0/8 is directly connected, Ethernet0/0  
L      21.0.0.2/32 is directly connected, Ethernet0/0  
S      30.0.0.0/8 [1/0] via 21.0.0.1  
S      162.14.0.0/16 [1/0] via 192.168.168.2  
      192.168.168.0/24 is variably subnetted, 2 subnets, 2 masks  
C      192.168.168.0/24 is directly connected, Ethernet0/1  
L      192.168.168.1/32 is directly connected, Ethernet0/1
```

```
R5#
```



R6



```
R6#sh ip route
```

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP  
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area  
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2  
        E1 - OSPF external type 1, E2 - OSPF external type 2  
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2  
        ia - IS-IS inter area, * - candidate default, U - per-user static route  
        o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP  
        a - application route  
        + - replicated route, % - next hop override, p - overrides from PfR
```

```
Gateway of last resort is not set
```

```
S      10.0.0.0/8 [1/0] via 192.168.168.1  
S      20.0.0.0/8 [1/0] via 192.168.168.1  
S      21.0.0.0/8 [1/0] via 192.168.168.1  
S      30.0.0.0/8 [1/0] via 192.168.168.1  
      162.14.0.0/16 is variably subnetted, 2 subnets, 2 masks  
C      162.14.0.0/16 is directly connected, Ethernet0/0  
L      162.14.0.1/32 is directly connected, Ethernet0/0  
      192.168.168.0/24 is variably subnetted, 2 subnets, 2 masks  
C      192.168.168.0/24 is directly connected, Ethernet0/1  
L      192.168.168.2/32 is directly connected, Ethernet0/1
```